

Automated Conveyor Belt System with Sensors

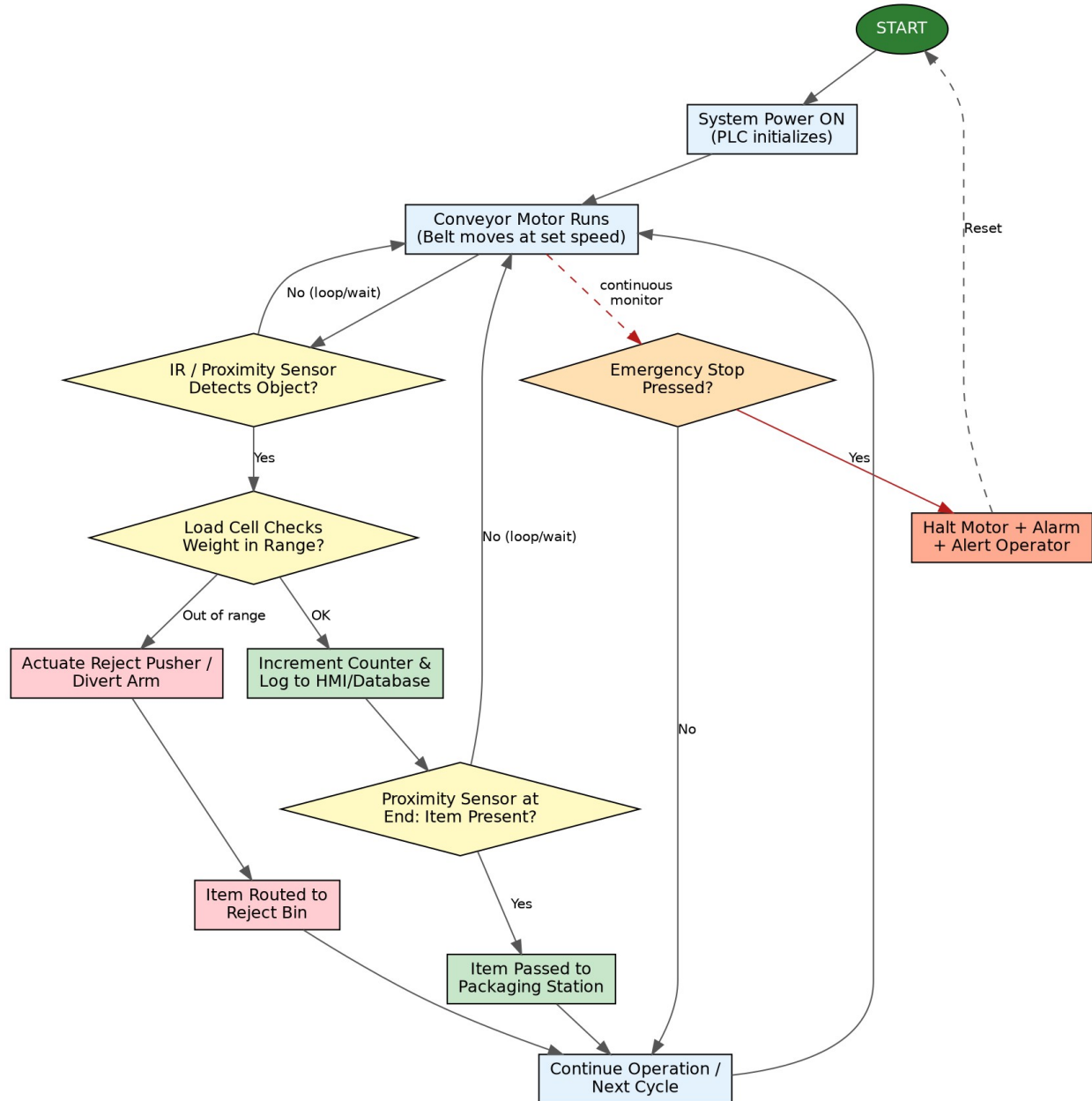
Task 3: Automation System Design — Flowchart & Explanation Report

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1. System Overview

This design automates the inspection, weight-verification, counting, and sorting of items moving along a factory conveyor line. A Programmable Logic Controller (PLC) coordinates the belt motor, an infrared (IR) proximity sensor at the entry point, a load cell for weight checking, a reject actuator for out-of-spec items, an exit proximity sensor, an emergency stop circuit, and a Human-Machine Interface (HMI) that logs throughput. The goal is to move items from intake to a packaging station automatically, diverting any item that fails a weight check, without requiring a human operator to inspect each item manually.

2. System Flowchart



3. Component Breakdown

3.1 Sensing

- Entry IR/proximity sensor — detects when an item enters the inspection zone and triggers the weight-check sequence.
- Load cell — measures item weight against a pre-set acceptable range (e.g., 495–505 g for a 500 g nominal product).

- Exit proximity sensor — confirms an accepted item has physically left the belt and reached the packaging station before the next cycle begins.

3.2 Control

- PLC — central controller that reads all sensor inputs, runs the decision logic, and drives the motor, reject actuator, and alarm outputs.
- HMI/database logging — records item count, reject count, and timestamps for traceability and performance reporting.

3.3 Actuation

- Conveyor motor — drives the belt at a fixed, controlled speed.
- Reject pusher/divert arm — a pneumatic or solenoid-driven actuator that pushes an out-of-range item off the main line into a reject bin.

3.4 Safety

- Emergency stop (E-stop) — a hardwired, normally-closed circuit that is monitored continuously and independently of the main PLC scan cycle, immediately halting the motor and sounding an alarm if pressed.

4. Logic Sequence

On power-up, the PLC initializes and starts the conveyor motor at a constant speed. As each item passes the entry sensor, the system checks the load cell reading against the acceptable weight window. Items within range are counted, logged to the HMI, and allowed to continue to the exit sensor, which confirms hand-off to the packaging station before the next item is processed. Items outside the acceptable range trigger the reject actuator, diverting them into a reject bin, and the cycle resumes. Throughout operation, the E-stop circuit is monitored independently of the main sequence; if pressed at any point, the motor halts immediately and an alarm alerts the operator, regardless of what stage the current item is in.

5. Benefits of This Design

- Removes manual weight-checking, increasing line speed and consistency.
- Produces a digital record (count, rejects, timestamps) useful for quality control and traceability.
- Independent E-stop monitoring ensures the safety response is not delayed by the PLC's normal logic scan.
- Modular design: the reject-and-divert logic can be extended with additional checks (e.g., a vision camera for surface defects) without redesigning the whole system.

6. Conclusion

This conveyor-with-sensors design demonstrates a practical, scalable pattern used widely in Industry 4.0 packaging and quality-control lines: sense, decide, act, and log, with a hardwired safety layer running in parallel. The same PLC-and-sensor architecture generalizes to smart home lighting or full automated packaging systems by swapping the specific sensors and actuators while keeping the same sense-decide-act-log logic.